Generate Document

AlgoExpert · Easy · Strings

The problem

- Can `document` be built from the multiset of characters in `characters`?

Intuition

- We can build the document if, for every character, we have at least as many copies available as needed.

Approach

- 1. Count available characters
- 2. For each document character, decrement its count
- 3. If any count goes negative, return false

Java solution

```java
public class GenerateDocument {
    public static boolean generateDocument(String characters, String document) {
        Map<Character, Integer> counts = new HashMap<>();
        for (char c : characters.toCharArray()) counts.merge(c, 1, Integer::sum);
        for (char c : document.toCharArray()) {
            int left = counts.getOrDefault(c, 0);
            if (left == 0) return false;
            counts.put(c, left - 1);
        }
        return true;
    }
}
```

Complexity

- Time: O(n + m)
- Space: O(c)

Recap

- We can build the document if, for every character, we have at least as many copies available as needed.
- Complexity: O(n + m)